

Ahmed Idriss

DATA & AUTOMATION ENGINEER · AI · AWS · PYTHON · DATA GOVERNANCE

SPECIALTIES

- Cloud data infrastructure & ETL pipelines
- Governed, auditable data (GDPR for health data)
- Automated reporting & stakeholder dashboards
- AWS data services at scale
- Workflow & process automation
- LLM integration & agentic workflows

CORE STACK

AI & AGENTS

LLM integration OpenAI
AWS Bedrock Agentic workflows
Multi-agent orchestration
Tool / function calling RAG
Prompt engineering LangChain
Human-in-the-loop Guardrails
PII redaction

DATA ENGINEERING

AWS RDS S3 Lambda Glue
Redshift EC2 PostgreSQL
MySQL MongoDB
Apache Airflow

AUTOMATION & ETL

Python Pandas SQLAlchemy
REST APIs Zapier
Custom workflow automation

REPORTING & BI

Power BI Tableau
Automated report pipelines
Dashboards Infographics

DATA GOVERNANCE

GDPR compliance Data quality
Access control Audit trails
Retention policies Documentation

PROGRAMMING

Python SQL JavaScript Bash
R

PROFILE

Results-driven Data Manager and Engineer with **5+ years designing, building and managing complex data infrastructure** across health technology, investment and commercial sectors. I deliver **automated reporting pipelines**, robust database architecture and **rigorous data governance**, including GDPR compliance for sensitive health data at scale. Experienced in deploying mobile data collection tools and turning complex data into clear outputs for senior stakeholders. First-class Mathematics & Computer Science graduate from the University of Bristol, with deep roots in and first-hand knowledge of East Africa.

5+

years in data engineering & management

1M+

users on the platform data I manage

3

sectors: health tech, investment, commercial

1st

Class Honours, University of Bristol

PROFESSIONAL EXPERIENCE

Data & Automation Engineer — Spectrum.Life

JUL 2022 – PRESENT

Dublin, Ireland · Health technology platform serving millions of users

AWS RDS S3 Lambda Redshift Python SQL GDPR Power BI

- Architected and managed the **data infrastructure for a platform serving millions of users**: relational databases, ETL pipelines and AWS cloud services (RDS, S3, Lambda, Redshift).
- Implemented an **organisation-wide GDPR compliance policy for sensitive health data**, covering data minimisation, access controls, retention policies and audit logging.
- Built **automated client reporting pipelines** for large insurance clients, producing timely, accurate, stakeholder-ready outputs.
- Designed and deployed **automated file processing workflows** for client onboarding, significantly reducing manual effort and error rates.
- Produced dashboards, infographics and management information packs that translate complex data into clear visuals for non-technical audiences.
- Trained and mentored colleagues on data systems, quality assurance and best practice, building team capacity.
- Worked in an Agile/Scrum environment, managing competing priorities across multiple client accounts with strict deadlines.

IT Data Engineer — WW Group Furniture

JUL 2021 – JUL 2022

Bristol, UK · Commercial / retail operations

Python ERP integration ETL SQL KPI dashboards

- **Automated a significant volume of sales order processing and stock purchasing workflows**, substantially reducing manual processing time and operational errors.
- Developed and maintained **data pipelines integrating ERP, inventory and sales systems** for real-time stock and order visibility.
- Produced management reporting and KPI dashboards for operational decision-making.

SECURITY & GOVERNANCE

- GDPR
- Data minimisation
- Access control
- Audit logging
- Retention policies
- Data quality frameworks
- Documentation standards

WAYS OF WORKING

- Agile / Scrum
- Git
- Jira
- Confluence
- Mentoring

HOW I WORK

- Sensitive data is protected by design, not as an afterthought.
- Every change is traceable: who accessed it, what ran, what changed.
- Automate the repetitive so people can focus on decisions.
- Reports should be clear to the person reading them, not just the analyst.
- Build team capacity: document, train and mentor.

LANGUAGES

- English — Native
- Dutch — Expert
- Somali — Expert
- French — Intermediate
- Arabic — Basic

PROFESSIONAL EXPERIENCE (CONT.)

Junior Data Engineer — Mbuyu Capital

AUG 2020 – JUL 2021

London, UK · Africa-focused investment

Database design SQL Python Financial reporting Data quality

- **Built the core investment database and reporting infrastructure from scratch**, covering multi-million dollar portfolio positions across African markets.
- Developed **automated reporting tools** giving portfolio managers accurate, timely financial data and performance analytics.
- Gained extensive exposure to data flows, quality control and reporting standards in a high-stakes financial environment.

FEATURED PROJECTS

DataEngine — AI Data-Operations Copilot

2026

Lead developer · For accounting firms and retailers: messy spreadsheets in, approved fixes and client-ready reports out.

Next.js 16 React 19 TypeScript Supabase PostgreSQL Python Polars DuckDB Docker GitHub Actions

- AI profiles client workbooks and proposes **materiality-ranked fixes**; an accountant approves each change before it becomes a new immutable version.
- **Queue-based agent architecture**: Postgres job queue with leases, so the agent host needs no inbound ports and a restart delays work instead of losing it.
- Deterministic maths in Polars and DuckDB; the LLM only makes judgment calls, through a **redaction layer** that keeps raw customer rows away from the model.
- Secured a **multi-tenant Supabase backend**: Row-Level Security, membership-checking database functions, tenant-scoped storage and Vault-held credentials.
- QuickBooks Online, Odoo and spreadsheet connectors; reports exported as PDF, Word, Excel and Markdown.
- CI/CD with GitHub Actions, Docker builds deployed over a private Tailscale network; 600+ Python tests plus an end-to-end suite.

NOVA — Governance Platform for AI Agents

2026

Lead developer · Run AI agents in a business under policy, with human approval and a verifiable audit trail.

Python React 19 TypeScript Tailwind 4 SQLite Docker Terraform AWS Bedrock

- Built a **multi-tenant control plane** over the open-source Hermes Agent runtime, without forking it.
- Compiled agent permissions into a **fail-closed pre-tool-call hook**, so a denied action is blocked inside the agent process, not just hidden in the UI.
- Designed a **write-ahead audit log**: intent before every change, committed or failed after, with the human actor, tenant and correlation ID. White-label Control Center (React 19) with RBAC, CSRF protection and strict CSP.
- Tenant isolation in SQL with fair scheduling and per-tenant concurrency limits; **human approval gates** and agent routing across Telegram, Slack, Discord, WhatsApp and Signal.
- **Terraform for AWS** (EC2, EBS, KMS, IAM, ECR, CloudWatch, SSM) with no inbound ports, IMDSv2 and per-integration IAM roles; 900+ automated tests.

EDUCATION

BSc Mathematics & Computer Science — First Class Honours

University of Bristol · Bristol, UK · Graduated 2020